One topic two learning designs

Jonathan Pincher | Educational design portfolio

Adapting a practical investigation of sticks for 15-year-old learners and first-year undergraduates

This portfolio demonstrates how I apply differentiation to meet the needs of different learning groups. I have presented two versions of the same topic: the teaching and learning pack for 15-year-olds and the first-year undergraduate teaching pack. Both documents are included for reference, alongside interactive demonstrations, to show how I adapt the level of challenge, scaffolding and assessment to each group's learning needs.

My approach

I start with the learning students need to demonstrate, then adapt challenge, support and assessment. The school version builds confident scientific enquiry; the undergraduate version develops quantitative modelling and critical judgement.

I redesign what learners investigate, how independently they work and the evidence and feedback they receive. Both versions connect objectives, activities and assessment, with accessible alternatives and explicit boundaries around AI use.

What this portfolio demonstrates

Design capability	Evidence in these examples
Understanding learning needs	Different assumptions about prior knowledge, mathematical preparation and independence.
Designing progression	Observation and fair testing develop into protocol design, model evaluation and qualified inference.
Aligning assessment	Age-appropriate outcomes are matched to tasks, marking criteria and individual evidence.
Building learning resources	Lesson plans, worksheets, visual aids and interactive browser demos support each route.
Using AI critically	AI assists drafting and critique; factual verification and teacher judgement remain explicit.

Explore the evidence

Open the original age 15 teaching pack

Open the original undergraduate teaching pack

These are design and prototype examples, not a completed classroom impact study. The interactive dashboards contain clearly labelled fictional learners; no attainment gains or preparation-time savings are claimed.

The key differences and why they matter

Design choice	Age 15 learners	First year undergraduates
Starting point	Mixed-attainment secondary learners developing scientific vocabulary, measurement and variables.	Basic algebra, SI units and graphing assumed; specialist beam theory introduced within the unit.
Learning demand	Observe, measure, plan a fair test, calculate means and explain a pattern.	Characterise specimens, design a protocol, estimate stiffness and evaluate a model.
Time and independence	Four 60-minute lessons with modelling, group roles and regular checks.	Four two-hour sessions, plus about six hours of independent reading, analysis and reporting.
Conceptual depth	Distinguish observation from inference and bending resistance from breaking strength.	Distinguish specimen stiffness, apparent modulus and strength; examine ideal-beam assumptions.
Use of data	Repeated displacement readings, means and a scatter graph of diameter against movement.	Force–deflection series, SI calculations, fitted gradients, residuals and uncertainty.
Support and extension	Vocabulary, sentence starters, structured tables and graph scaffolds; extensions explore sampling.	Algebra refreshers and scaffolded spreadsheets alongside open analysis; extensions address sensitivity and competing models.
Assessment evidence	Short report or recorded presentation; 15-mark rubric and separate 10-mark knowledge check.	1,500-word report with data and calculation appendices; suggested 90% report and 10% oral check.
AI engagement	Teacher-mediated resources and checked flawed examples; no student AI account required.	Optional approved AI critique or coding support with disclosure and independent verification; non-AI alternative retained.

These choices respond to the intended learning context, rather than treating age as a complete description of ability. Diagnostic tasks and feedback would determine how much support individual learners actually need. Curriculum and programme mappings remain provisional until checked against local requirements.

How the learning changes in practice

Example 1 From a fair comparison to a defensible model

For a 15-year-old, I ask: How does stick diameter relate to bending under the same load? Learners identify variables, repeat readings and explain why moisture or condition could affect their conclusion. The focus is a fair comparison and a reasoned account of the pattern.

For an undergraduate, I ask: What can force–deflection measurements tell us about specimen stiffness, and when is an apparent modulus defensible? Learners justify sampling, inspect the fitted relationship and evaluate geometry and model assumptions. The focus moves to the validity and limits of a quantitative explanation.

Example 2 From support prompts to justified decisions

The school worksheet helps learners organise observations and conclusions through vocabulary, recording tables and sentence starters. The undergraduate worksheet requires a traceable protocol, dimensional consistency and a rationale for analysis decisions. Both provide access, while the expected reasoning changes.

Example 3 From checking an answer to interpreting evidence

The age 15 interactive activity checks means and comparisons, then asks learners to challenge the statement that the thickest stick is always strongest. The undergraduate activity checks stiffness and apparent modulus calculations, then asks whether a strong statistical fit validates the physical model. Each interaction targets a misconception appropriate to its learning outcomes.

Interactive demonstrations

Launch the age 15 student lab and teacher dashboard

Launch the undergraduate student lab and teacher dashboard

Both demos include an animated bending model, answer boxes and a linked Submit to tutor action within the same browser. Fictional submissions load automatically so a reviewer can inspect the teacher view without entering data. The demos are companions to the fuller teaching packs, rather than complete replacements for practical work.

Objective practice questions are scored automatically. Written responses are gathered for tutor review against the relevant rubric. The displays help identify possible learning gaps; they do not establish teaching effectiveness or provide validated automated marking of extended reasoning.

My design judgement and evaluation approach

What stays consistent

Both versions retain a clear enquiry sequence, accessible resources, individual evidence of understanding and conclusions bounded by the data. I preserve these principles while changing the mathematical demand, degree of independence and assessment standard. Practical piloting and teacher-approved procedures remain essential at both levels.

What AI contributes

I used generative AI collaboratively to develop and iterate planning documents, differentiated activities, assessment prompts and prototype interfaces. I directed the audience changes and requested revisions to interaction, submission flow and teacher reporting. The technical implementation was AI-assisted. The portfolio makes the use of AI visible rather than presenting generated material as independently verified teaching evidence.

My intended workflow includes checking content, calculations, accessibility and alignment before delivery, then revising resources using actual learner evidence. The prototypes demonstrate features and scoring logic; classroom usability and impact still require piloting. A specific AI model comparison has not been conducted.

How I would evaluate the teaching

Evidence to collect	Design decision it would inform
Matched entry and exit responses	Whether key misconceptions changed and which explanations need revision.
Individual work and rubric patterns	Whether learners can demonstrate the intended outcomes independently.
Participation and accessibility feedback	Which supports enable access and where barriers remain.
Practical timings and data quality	Whether the investigation is feasible and instructions produce usable evidence.
Tutor feedback and moderation	Whether marking criteria are clear and interpretations consistent.

Reference and viewing instructions

Download and extract the complete portfolio ZIP before opening this document. Keep this document and its references folder together so the relative links can find the originals. Some PDF readers block local-file links; if prompted, open the corresponding file directly from the references folder. HTML demos open in Safari or Chrome. The single-browser demos do not collect submissions across separate devices.

The two supplied original Word teaching packs are included unchanged. This comparison summarises their design choices and identifies the later interactive prototypes separately. Suggested curriculum alignment, timings and assessment weights should be adapted to the employing institution.